Ben Rosin
3/15/26

Midterm Research Project: Numerical experiments on quantum relaxation

Introduction

I used Matlab to model the relaxation of a quantum state in two systems:
In system *A*, a state $|1\rangle$, of energy E_1, is coupled to a manifold of states $\{|k\rangle\}$ with coupling constants V_{1k}.
In system *B*, a state $|0\rangle$, of energy E_0, is coupled with coupling constant V_{01} to state $|1\rangle$, which remains coupled to the manifold of states $\{|k\rangle\}$ with coupling constants V_{1k}. In both states, I took V to be real, and thus $\{V_{1k}=V_{k1}^*\}$, and $\{V_{01}=V_{01}^*\}$.
I observed the effect of varying ΔE (the energy span of manifold $\{|k\rangle\}$), N_k (the number of states in the manifold $\{|k\rangle\}$), V_{10}, V_{1k}, E_0, and E_1.
My results, conducted with a finite N_k and ΔE, were compared to the theoretically derived model wherein N_k and ΔE are infinite.

Methods

Matlab code was written to mimic solving the time-dependent Schrodinger equation for Systems A and B:

$$\frac{\partial\psi(t)}{\partial t} = -\frac{i}{\hbar}H\psi \qquad (1)$$

Where the wavefunction and hamiltonian for System A is given by:

$$\psi(t) = C_1(t)\,|1> + \sum_{k=0}^{N_k} C_k(t)\,|k> \qquad (2)$$

$$H = E_1|1><1| + \sum_{k=0}^{N_k} E_k|k><k| + \sum_{k=0}^{N_k}(V_{1k}|1><k| + V_{k1}|k><1|) \qquad (3)$$

And the wave function and hamiltonian for System B is given by:

$$\psi(t) = C_0(t)\,|0> + C_1(t)\,|1> + \sum_{k=0}^{N_k} C_k(t)\,|k> \qquad (4)$$

$$H = E_1|0><0| + E_1|1><1| + V_{01}|0><1| + V_{10}|1><0| + \sum_{k=0}^{N_k} E_k|k><k| + \sum_{k=0}^{N_k}(V_{1k}|1><k| + V_{k1}|k><1|) \qquad (5)$$

Thus, for a given *C(t)*, rearranging equation (1) yields:

$$C(t) = e^{\frac{-i}{\hbar}Ht}C(0) \qquad (6)$$

And, by diagonalizing *H* and recombining *C(t)* into *Ψ(t)*:

$$\psi(t) = \sum_{j=0}^{N_k} e^{\frac{-i}{\hbar}E_j t}|j> <j|\psi(0)> \qquad (7)$$

Where E_j and $|j\rangle$are the eigenvalues and eigenvectors of H. Equation (7) can readily be evaluated numerically by MATLAB, given an initial $\Psi(0)$ and Hamiltonian. Given $\Psi(t)$, P_0, P_1, or P_k can be easily obtained via:

$$C(t) = < 1|\psi(t) > \quad (8)$$

$$P(t) = |C(t)|^2 \quad (9)$$

Also evaluated with Matlab.

It was observed that the inclusion of the reduced planck's constant drastically increased computation time and served no purpose for the comparison of Probabilities across the parameter space, and so the constant was normalized.

Theoretical Model

My results were compared to the theoretical model derived in class for Systems A and B. the theoretical model assumes the manifold $\{|k\rangle\}$ is infinitely large and spread across an infinite ΔE. This derivation makes use of a laplace transform to obtain $C_1(t)$ in terms of Green's Function:

$$C_1(t) = -\frac{1}{2\pi i}\int_{-\infty}^{\infty} dE e^{-iEt/\hbar} G_{1,1}(E + i\epsilon);\ \epsilon \to 0_+ \quad (10)$$

Where, using the Dyson Identities,

$$G_{1,1}(E) = \lim_{\epsilon\to 0} \frac{1}{E+i\epsilon-E_1-B_1(E)},\ B_1(E) = \sum_k^{\infty} \frac{|V_{1,k}|^2}{E - E_k + i\epsilon} \quad (11)$$

However, in the theoretical model, the infinite size of $\{|k\rangle\}$ allows for a substitution:

$$\sum_k^{\infty} \frac{|V_{1,k}|^2}{E - E_k + i\epsilon} \to \frac{1}{2\pi}\int_{-\infty}^{\infty} dE_k \frac{\Gamma_1(E_k)}{E - E_k + i\epsilon} \quad (12)$$

$$\Gamma_1 \equiv 2\pi\overline{|V_{1K}|^2 \rho_K(E_k)} \quad (13)$$

Where Γ_1 is defined as the average product of the coupling and density of states over all continuum states with energy E_k. Finally, because the manifold is of infinite size, the integral is solved by assuming Γ_1 is independent of E:

$$B_1(E) = \frac{\Gamma_1}{2\pi}\int_{\infty}^{\infty} \frac{1}{E-x+i\epsilon} = -\frac{1}{2} i\Gamma_1 \quad (14)$$

This form of $B_1(E)$ yields, from equations (10) and (11):

$$C_1(t) = C_1(0) e^{-iE_1 t/\hbar - \frac{1}{2}\Gamma_1 t/\hbar} \quad (15)$$

$$P_1(t) = |C_1(t)|^2 = e^{-\Gamma_1 t/\hbar} \quad (16)$$

Thus, in the theoretical model, $P_1(t)$ should display exponential decay (assuming $C_1(0) = 1$). Via the normalization of $\Psi(t)$, P_k should display a logarithmic growth.

Across Systems A and B, the analyses of the decay of the initial state's probability of being occupied are treated identically. The only difference is that the "manifold" in System B consists of state $|1\rangle$ as well as the manifold $\{|k\rangle\}$. The results obtained via Matlab were compared with the expected exponential decay of the probability of occupying the initial state and logarithmic growth of the probability of occupying the manifold in Systems A and B.

Results and Discussion

I plotted the probability of occupying each quantum state in systems A and B as a function of time. P_0 represents the probability of occupying $|0\rangle$, P_1 the probability of occupying $|1\rangle$, and P_k the probability of occupying any state $|k\rangle$ in the manifold $\{|k\rangle\}$.

In State *A,* each probability was initially plotted from $t = 0$ to $t = 4$. I observed that P_1 and P_k oscillated with a very high frequency *(figure 1)*

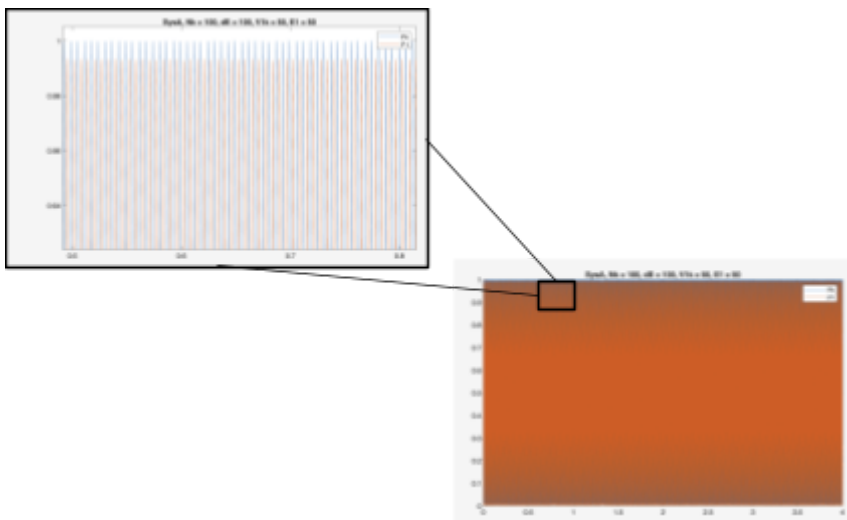

Figure 1: Pk and P1 plotted from t = 0 to t = 4. The parameters of the model are listed above the plot.

To allow for easier distinction between P_1 and P_k, I re-plotted each probability in system A from $t = 0$ to $t = 1$. Notably, *figure 1* provides no evidence of exponential decay in P_1 or growth in P_k. I explored the parameter space with subsequent experiments in search of agreement with the theoretical model.

I chose Nk from typical values as provided by professor Nitzan. Nk = 100 was chosen specifically to make my calculations as fast as possible while still falling within the bounds of a typical range. ΔE was chosen as 100 so that the density of states in the manifold $\{|k\rangle\}$ was $\rho = \frac{N_k}{\Delta E} = 1$. Thus, the spacing between states in the manifold $\varepsilon = \rho^{-1}$ was 1. Therefore,

choosing V_{1k} = 50 Satisfied the requirement that $\Delta E \ >> \ V_{1k} \ >> \ \rho^{-1}$, necessary so that the transition probability between $|1\rangle$ and $\{|k\rangle\}$ is non negligible, and so the occupation of one state can be distinguished from occupation of the other. To investigate the effects on the model when this requirement is violated, I ran an experiment with *ΔE* = 1000 and V_{1k} = 10 *(figure 2)*, and another with *ΔE* = 100 and V_{1k} = 100 *(figure 3)*. To ensure that E_1 remained in the center of the manifold's energy range, E_1 was increased to 500 when *ΔE* = 1000.

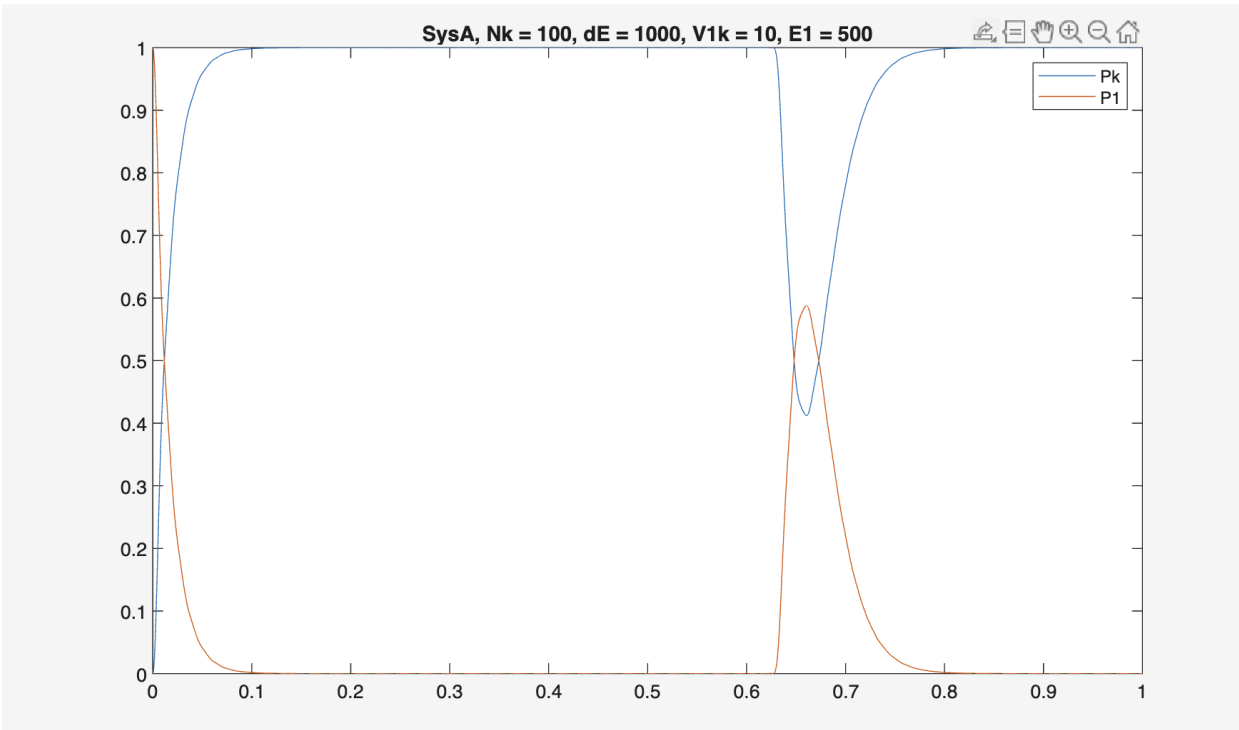

Figure 2: P_1 and P_k wherein V_{1k} = ρ^{-1}. Other parameters used are listed above the plot.

As shown in *Figure 2*, a coupling constant outside of the range $\Delta E \ >> \ V_{1k} \ >> \ \rho^{-1}$ causes the minor oscillations within P_1 and P_k to disappear, drastically lowering the probability of $|1\rangle$ being occupied at any given *t*.

Additionally, the hump in P_1 and dip at various P_k at *t = 0.6,* which appear in *Figures 5, and 6* and as oscillations in *figures 12, and 13*, are likely an artifact introduced by the small number of states in $\{|k\rangle\}$, and thus small ρ, relative to *ΔE*. These fluctuations in P_1 and P_k may be due to the difficulty in transitioning through $|1\rangle$between given $|k\rangle$states in the manifold, as each state no longer satisfies the requirement $V >> \rho^{-1}$.
This limitation conflicts with the theoretical model, which demonstrates a uniform, higher order exponential decay of P_1.
I hypothesize that these parameters offer a poor reflection of a Quantum Mechanical system relaxing as it eliminates the possibility of a transition between $\{|k\rangle\}$ and $|1\rangle$at many given *t*

values, along with a higher order decay of P_1 and growth of P_k, in favor for sudden changes in P_1 and P_k at specific, infrequent values of *t*.

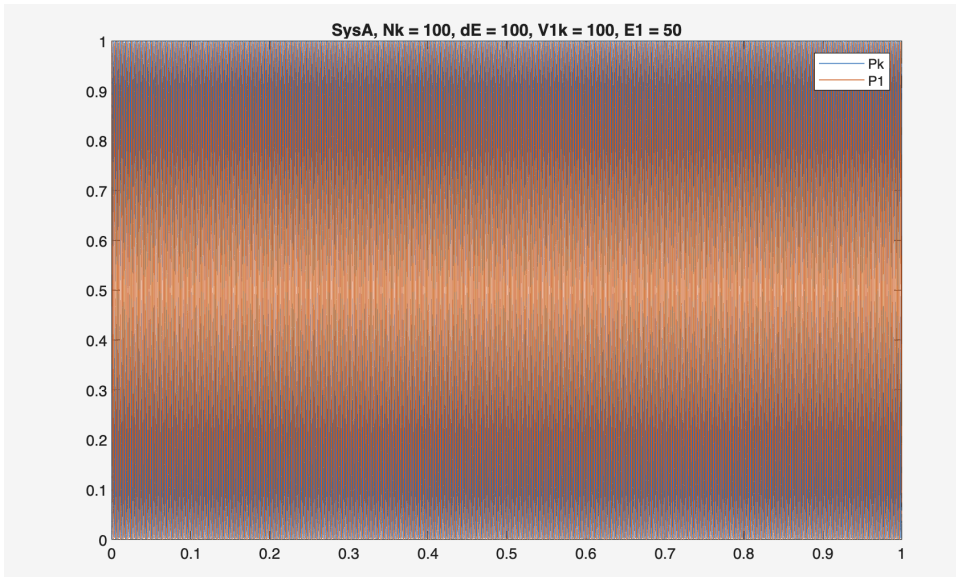

Figure 3: P_1 and P_k wherein $V_{1k} = \Delta E$. Parameters used are listed above the plot.

Conversely, *figure 3* demonstrates a coupling constant too large in magnitude. Here, the occupation of state $|1\rangle$ is indistinguishable from that of the manifold $\{|k\rangle\}$, which is not reflective of the Quantum Mechanical system. It is also not especially useful for analysis, as no dominant lineshape can be identified.

To investigate the effect of varying V_{1k} within the prescribed range, I Took V_{1k} to be 25 and 75 *(Figure. 4)*

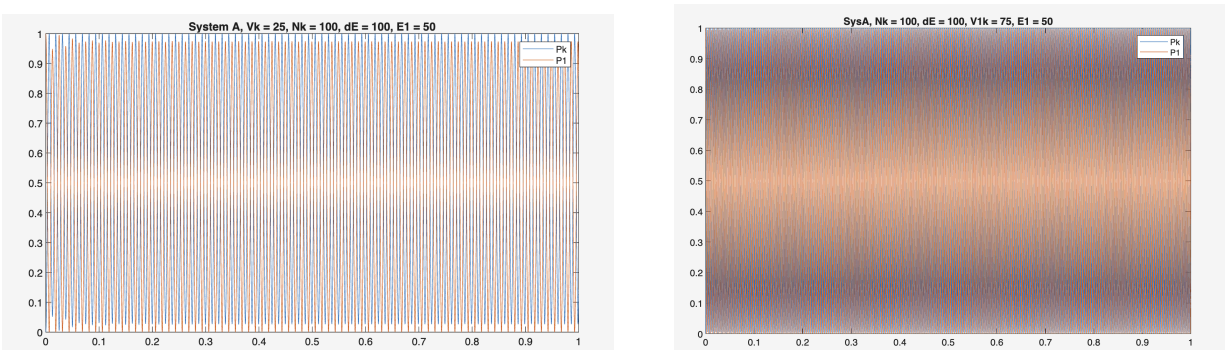

Figure 4: Varying V_{1k}. Parameters used are listed above each plot.

As visible from figures 1–4, a higher coupling constant increases the frequency of oscillations in P_1 and P_k, indicating a more frequent transition of quantum states. T However, with ΔE so close in magnitude to V_{10}, no higher order decay is observed in *figures 3 or 4*, as occupation of the initial state and manifold cannot be readily distinguished.

Without varying V_{10}, I investigated the effect of varying the energy spacing between manifold states, by increasing ΔE to 1000 *(figure 5)*. To ensure that E_1 remained in the center of the manifold's energy range, E_1 was increased to 500.

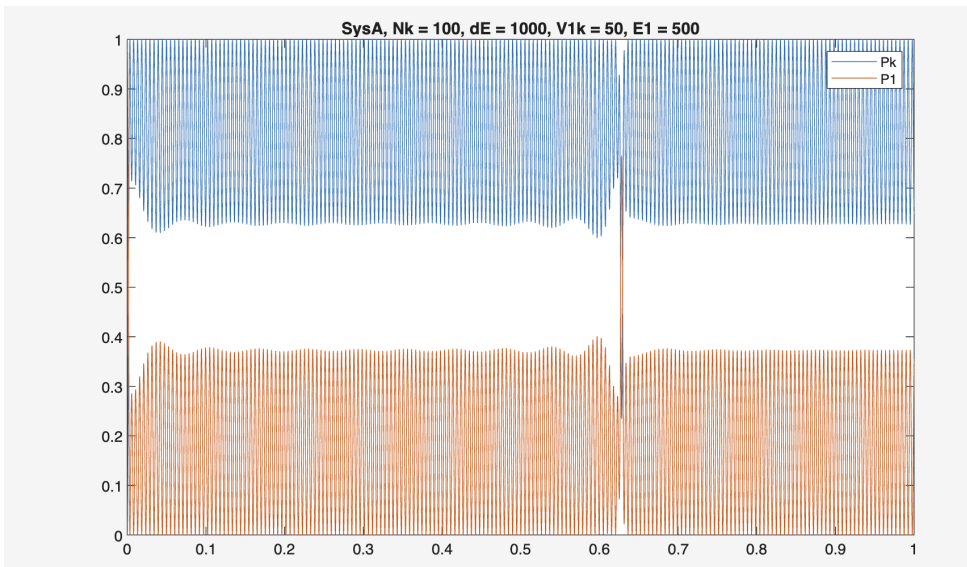

Figure 5: Increasing ΔE and E_1. Parameters used are listed above each plot.

From Figures 2 and 5, we see that increasing ΔE, and thus decreasing ρ, causes a decay in P_1 and growth in P_k when E_1 increases with ΔE. When E_1 is left at 50 *(Figure 6)*, I observed the dependence on the difference in probabilities P_1 and P_k on E_1 vs ΔE: A larger ratio of ΔE to E_1 allows for a higher probability P_1.

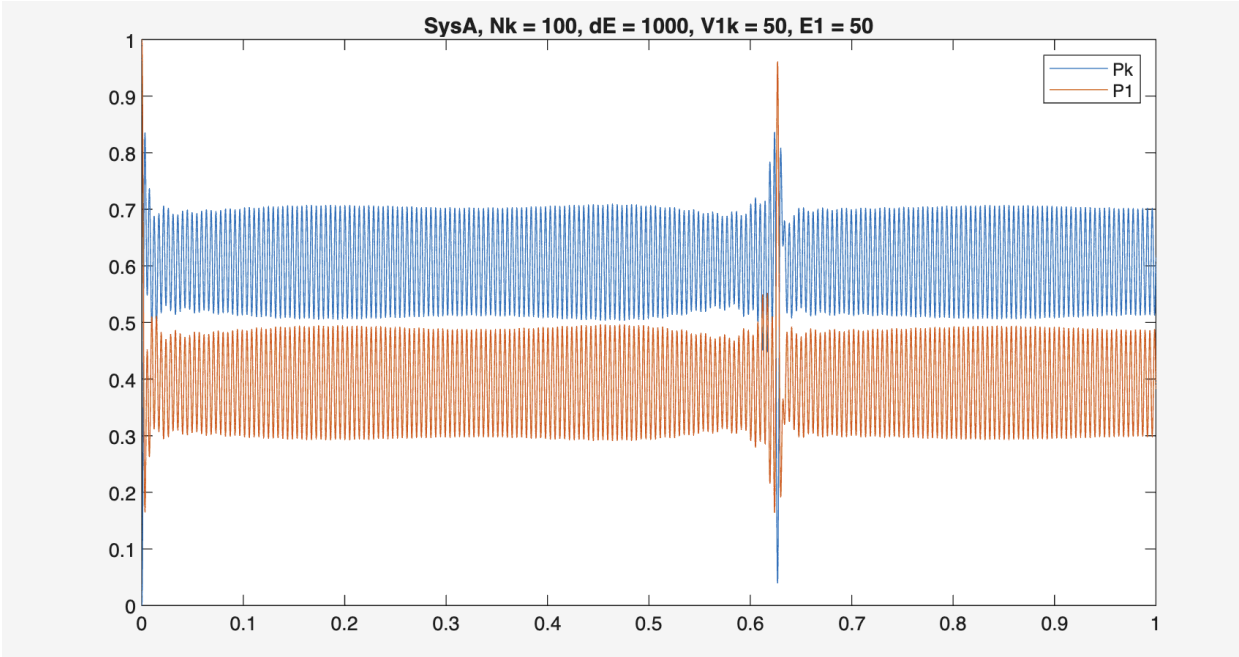

Figure 6: Increasing ΔE alone. Parameters used are listed above the plot.

However, when *ΔE* is left at 100 but E_1 is increased to 90 *(Figure 7)*, I observed similar probabilities as those displayed by *Figure 4*. This might indicate that the difference in P_k in P_1 is largely dependent on the size of *ΔE*, and not on E_1. The lack of decay in P_k and growth in P_1 in *Figure 7*, and presence thereof in *Figures 5 and 6*, indicates that the artificially small *ΔE* is limiting the separation of P_1 and P_k. In the theoretical model, wherein $\Delta E = \infty$, P_1 and P_k show exponential decay and logarithmic growth. An infinite *ΔE* is thus essential to the theoretically obtained lineshapes. However, the minor oscillations visible in P_1 and P_k in *Figures 5 and 6* may agree with the theoretical coupling of two individual states, although they do not match our theoretical model of coupling to an infinite manifold.

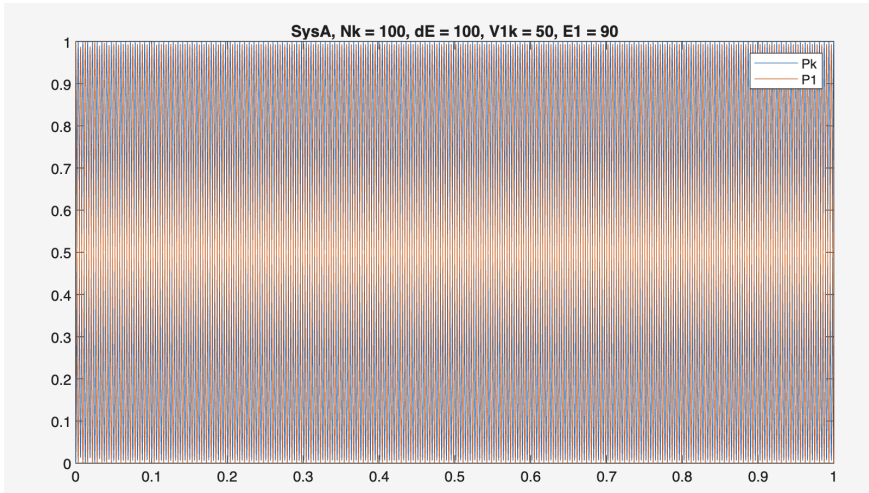

Figure 7: Increasing E_1 alone. Parameters used are listed above the plot.

Finally, to investigate the impacts of a manifold consisting of fewer $|k\rangle$states, I lowered N_k *(Figure 8)*. I found that lowering N_k decreased the frequency of the oscillations, but didn't appear to affect the relative probabilities of the state $|1\rangle$or manifold $\{|k\rangle\}$ being occupied. I predict that, as our in-class model assumed that the manifold consists of infinite states over an infinite energy spacing, this decrease in the oscillations of frequency is a departure from the P_1 and P_k given for an infinite manifold.

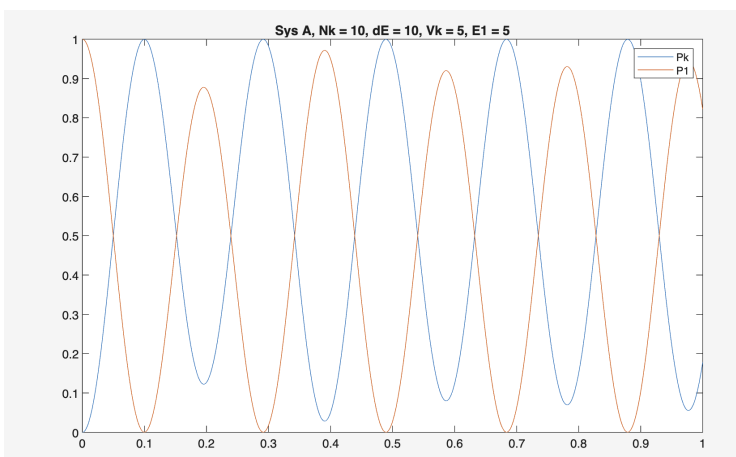

Figure 8: Decreasing N_k. V_{k1}, E_1 and ΔE were adjusted proportionally. Parameters used are listed above the plot.

In investigating System B, I considered the relative values of V_{10} and V_{1k}, and the relative values of E_1 E_0. Initially, I set standard parameters as described in *Figure 9*. The inclusion of the intermediary state $|1\rangle$between $|0\rangle$and the manifold $\{|k\rangle\}$ leads to a decay the probability of occupying the initially occupied $|0\rangle$, and a growth in P_k. P_0 remains nearly negligible in all of my experiments. In addition to their overall growth/decay, which agrees with the theoretical model P_0, P_1 and P_k display minor oscillatory behavior in each modelling of System B, which can be expected of two individual coupled states.

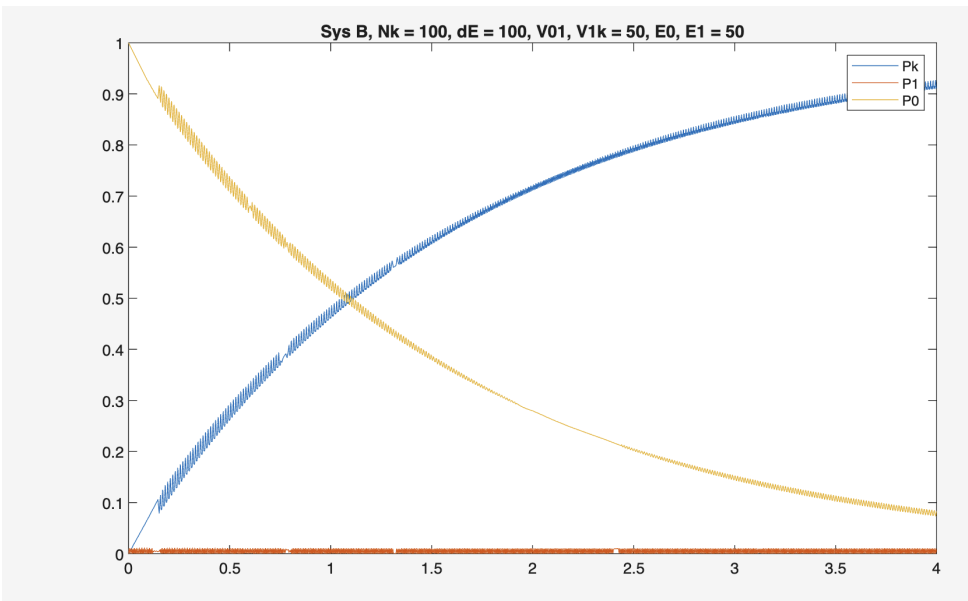

Figure 9: P_0, P_1 and P_k in System B. System parameters used are listed above the plot.

I began by decreasing V_{01} so that $V_{01} < V_{1k}$ and by decreasing V_{1k} so that $V_{1k} < V_{01}$ *(Figure 10)*. Interestingly, the adjustment of V_{01} and V_{1k} shifted the point at which $P_k = P_0$, increasing or decreasing the rate of change of the two probabilities when V_{1k} or V_{01} were larger, respectively. I predict this dependence on the coupling constants extends to the theoretical model.

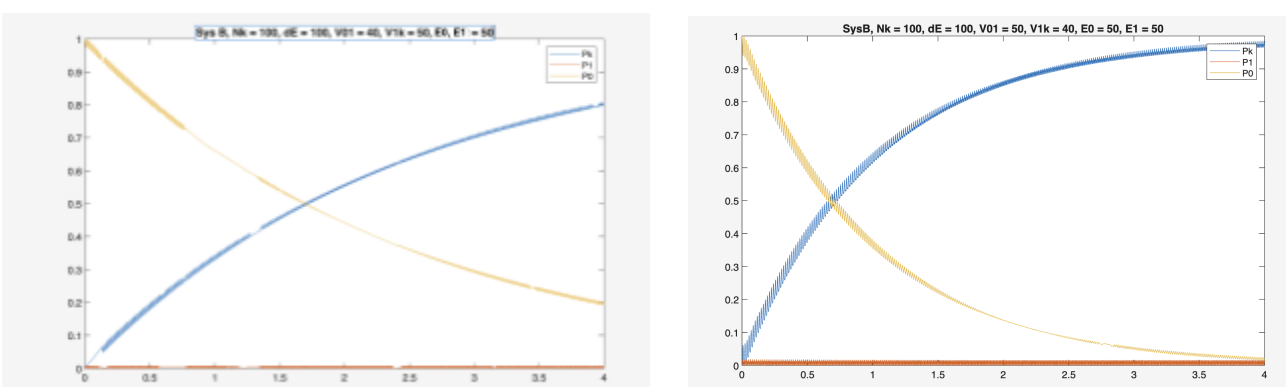

Figure 10: Decreasing the values of V_{01} (Left) and V_{1k} (Right). System parameters used are listed above both plots.

I then investigated the difference in E_0 and E_1 *(Figure 11)* by changing the value of E_1 relative to E_0, which I held at 50. Interestingly, I didn't see a large difference in the overall decay/growth of P_k and P_0, although the frequencies of each probability's oscillations changed with E_0-E_1.

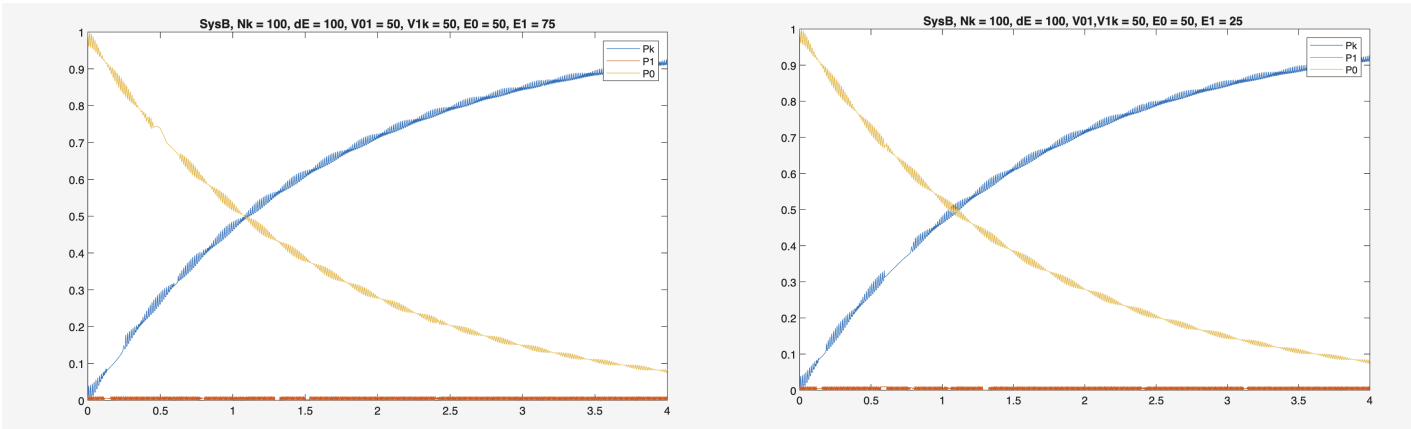

Figure 11: Creating nonzero (negative, left and positive, right) E_0-E_1. System parameters used are listed above both plots.

However, increasing *ΔE*, along with E_0 and E_1 proportionally, causes a change in the behavior of P_0 and P_k *(figure 12)*. Although P_0 and P_k still undergo frequent, small amplitude oscillations, a higher order oscillation has been introduced to their behavior. This plot is reminiscent of system A's behavior under similar parameters *(figure 2)*. Again, I believe this is due to the fact $N_k, \rho << \Delta E, V$. Notably, under these parameters, P_1 occasionally reaches non-negligible values, indicating that state $|1\rangle$spends more time occupied when *V* is not greater than ρ^{-1}. This behavior further demonstrates the difficulty in transitioning back and forth between the initial and manifold states, which introduces the observed disagreement with the theoretical model.

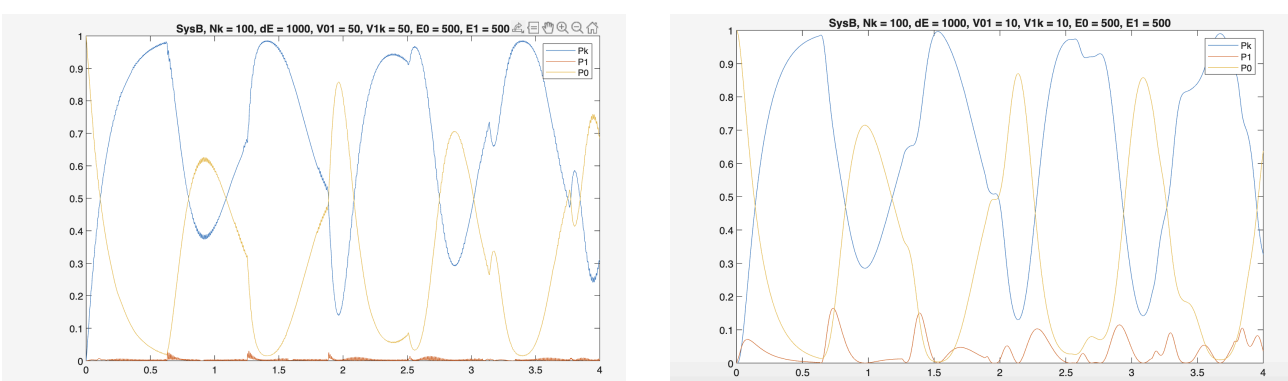

Figure 12: Increasing ΔE with varying V. System parameters used are listed above both plots.

Notably, increasing *ΔE* without adjusting E_0 and E_1 *(figure 13)* only affects the minor oscillations of the Probabilities–the higher order shape of the probabilities are identical to those observed in *figures 12, 2, 5 and 6*.

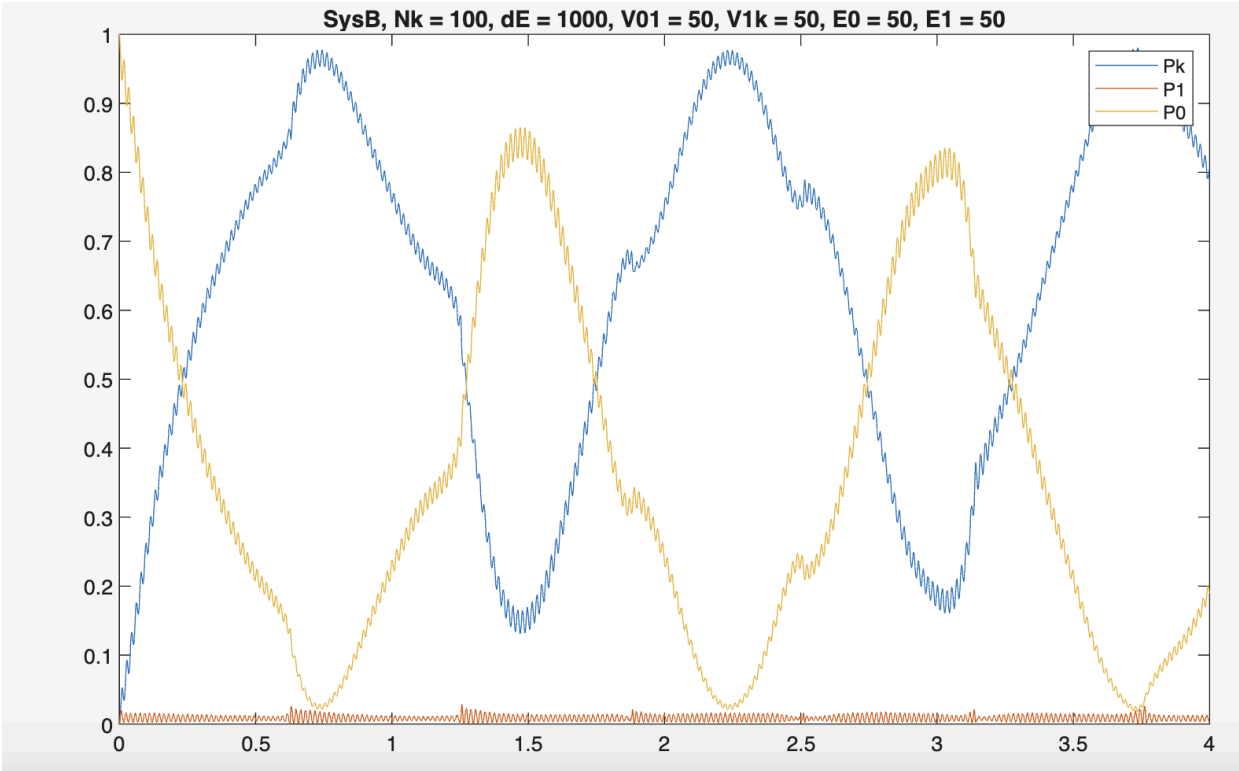

Figure 13: Increasing ΔE without adjusting E_0 and E_1. System parameters used are listed above each plot.

Similarly, when the coupling constants are set equal to *ΔE (figure 14)*, the oscillations within P_0, P_1 and P_k gain a very high frequency, but no other attributes of the plot change. This again indicates that the higher order oscillations visible in *figure 12* are due to the absolute magnitude of *ΔE*, not its magnitude relative to *V*. This increased frequency in minor oscillations with *V* agrees with theoretical predictions for two coupled quantum mechanical states, as a higher degree of coupling should facilitate transitions between the two states.

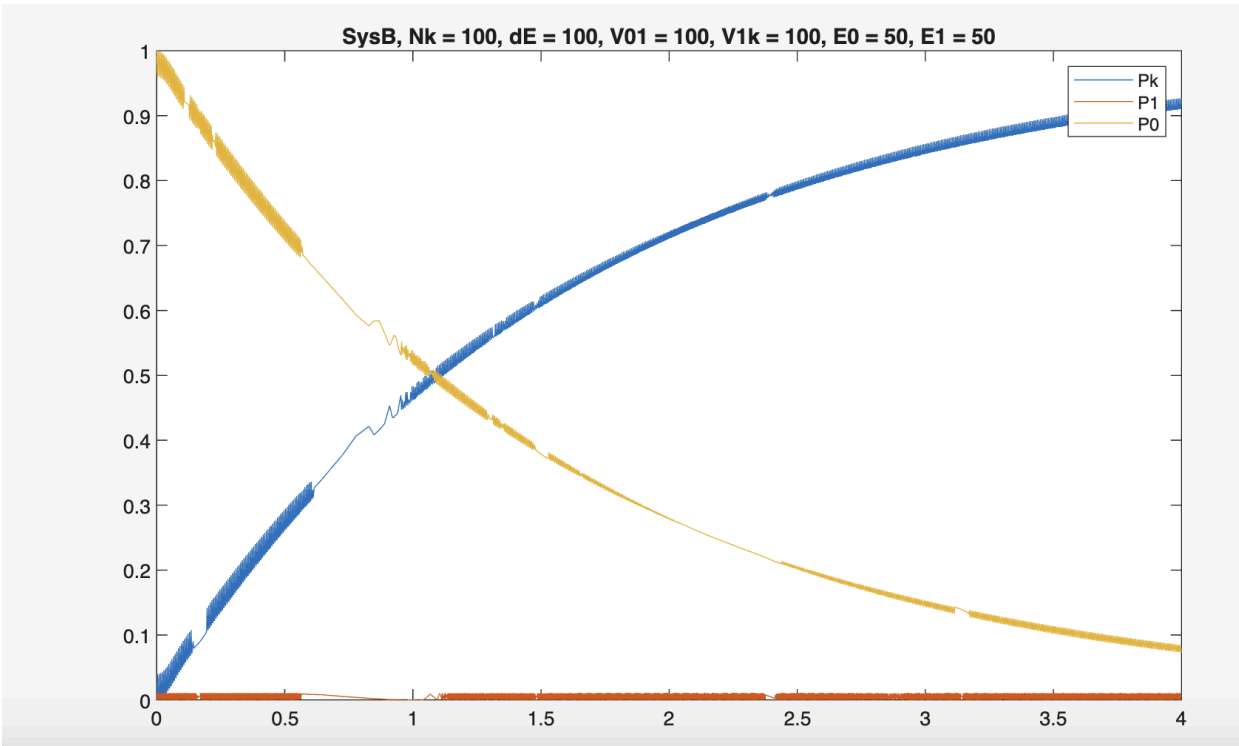

Figure 14: Increasing V. System parameters used are listed above the plot.

Finally, A lower N_k *(figure 15)* resulted in lower frequency oscillations within the P_0, P_1, and P_k, but did not change the overall shape of the graph. This behavior is akin to the behavior observed in system A *(figure 8)*. Notably, for a theoretically derived model, wherein $N_k = \infty$, these oscillations would be infinitely fine, causing individual peaks to be indistinguishable against the overall lineshape, giving rise to the theoretical lineshapes. This is a notable difference between the theoretical model and my own.

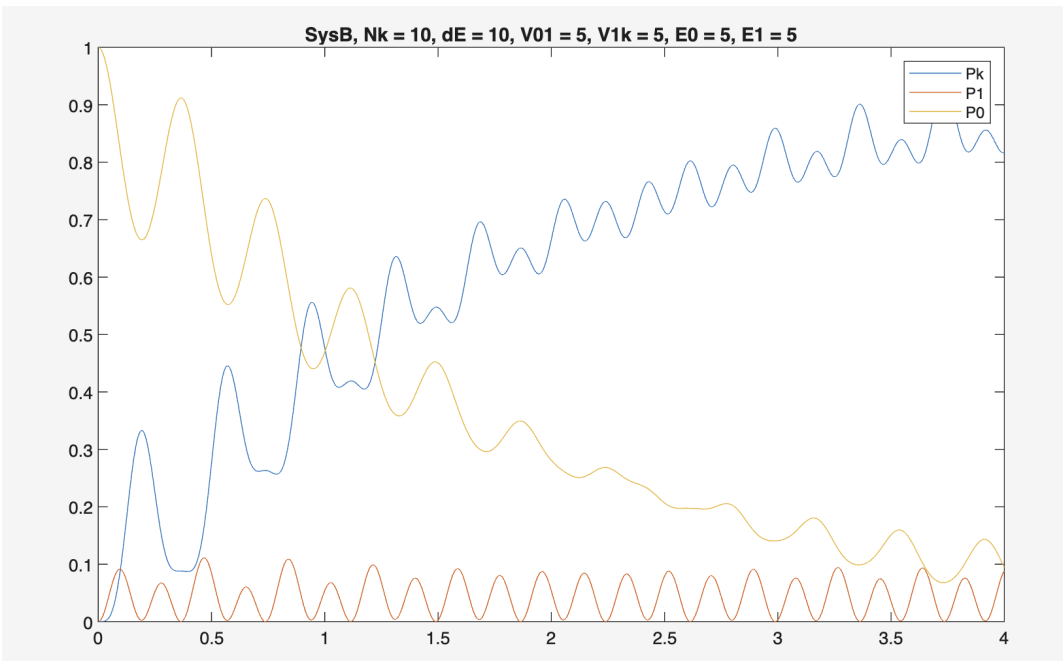

Figure 15: Lowering N_k. System parameters used are listed above the plot.

I then revisited systems A and B using a random distribution of the elements V_{1k}. I randomly assigned +/- signs to the elements of the vectors describing coupling to the manifold $\{|k\rangle\}$ *(figures 16, 17)*. Randomizing the signs had no effect on the resultant plots of P_0, P_1 and P_k. I re-investigated each system presented in *figures 1-16* with randomized V_{1k} elements and found no changes to the probabilities of occupying any given state.

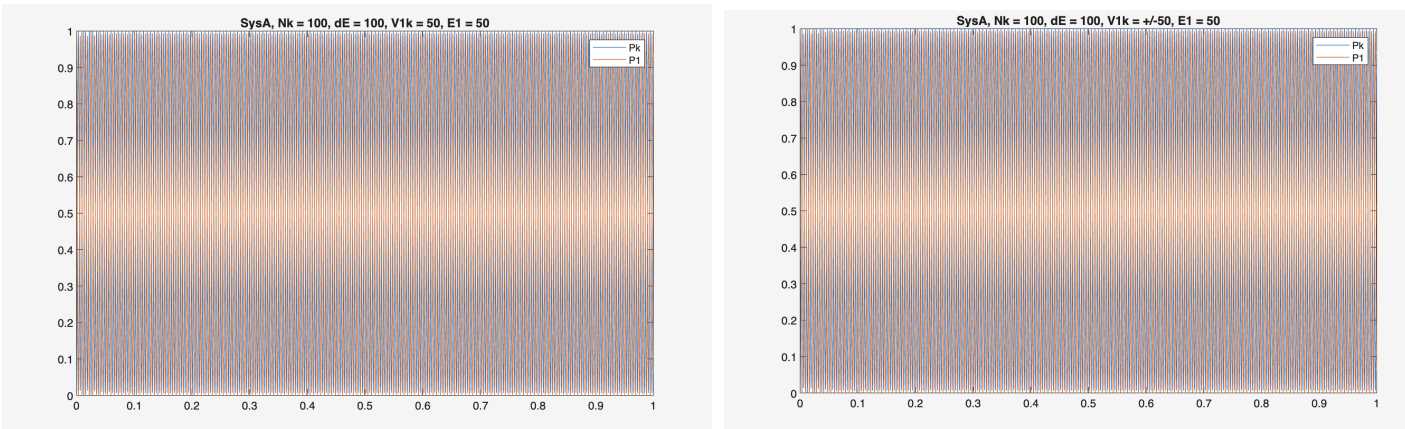

Figure 16: Uniform (left) and Random-sign (right) elements of the V_{1k} vector in System A. System parameters used are listed above each plot.

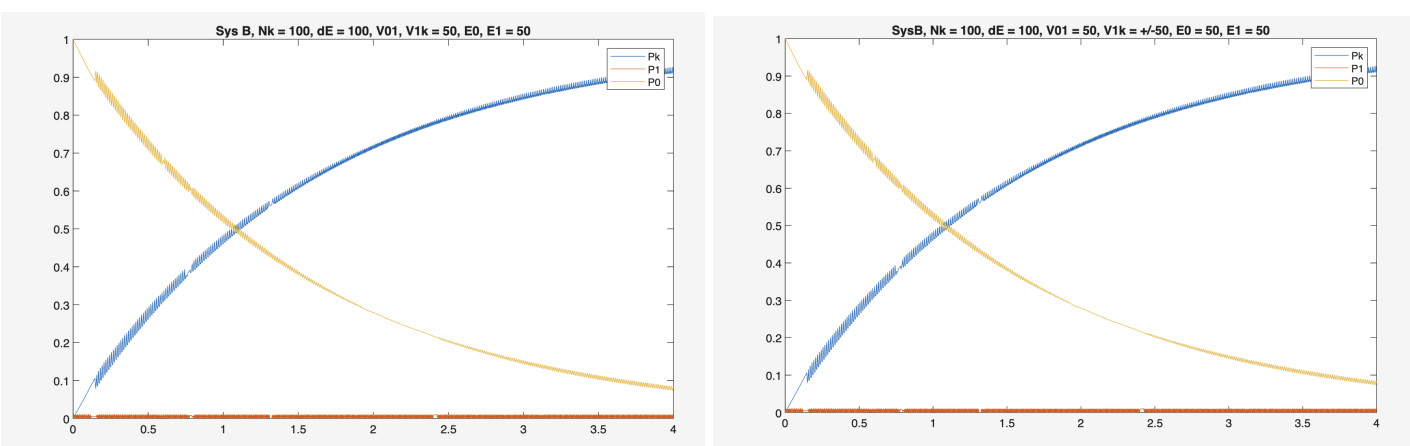

Figure 17: Uniform (left) and Random-sign (right) elements of the V_{1k} vector in System B. System parameters used are listed above each plot.

Conclusion

The purpose of this experiment was to investigate the time evolution of Quantum Mechanical relaxation in Systems A and B. Notably, this model included a finite manifold $\{|k\rangle\}$, which caused deviation from the theoretically derived behavior of both systems, which assumes manifold of infinite size. It found that limiting N_k caused visible minor oscillations within the lineshapes defining P_1, P_0, and P_k. The coupling constant between states was proportional to the frequency of these oscillations, and a coupling constant equal to the energy spacing ρ^{-1} between states in $\{|k\rangle\}$ resulted in the absence of these oscillations entirely, and introduced higher order fluctuations and oscillations in the lineshapes. A coupling constant too close to ΔE caused the probability of the initial state being occupied and probability of the manifold being occupied to become indistinguishable in System A, but did not have the same effect in System B I hypothesize that state $|1\rangle$, acting as a "doorway" between the initial state $|0\rangle$and the manifold, may have facilitated a clearer distinction between $|0\rangle$and $\{|k\rangle\}$, allowing P_1 and P_k to remain distinguished..

I discovered that this "doorway state" $|1\rangle$ was negligibly occupied, save for when the coupling constants between states were especially low relative to ΔE. I also noted that increasing ΔE in System A procured some decay in P_1 and growth in P_k, although the resultant plots were still limited in accuracy due to the finite size of the manifold. These results indicate that deviation from the theoretical lineshapes is due to the artificially low ΔE, which is taken to be infinite in the theoretical model. Similarly, a finite ΔE in System A may be the cause of a lack of observed decay in P_1 or growth in P_k.

However, increasing ΔE tenfold without increasing N_k caused another deviation from the theoretical model: higher order oscillations in the probabilities of the initial and manifold states. I theorize this to be due to the large energy spacing between energy states, hindering back-and-forth between the initial state and the manifold.

Higher V in both systems caused higher frequency in lower-order oscillations, which agrees with the theoretical prediction for two coupled quantum states–a higher degree of coupling facilitates transitions between the two states. This behavior, typical of individual coupled states, is likely introduced by the finite nature of this model.

Ultimately, the presence of individual peaks within the growth and decay lineshapes of the initial and manifold probabilities is a marked disagreement with the theoretical model, which demonstrates no distinguishable peaks when $N_k = \infty$. When N_k is lowered, the peaks are further distinguished, indicating that the assumption that $N_k = \infty$ allows for the elimination of these lower order oscillations.

Finally, randomizing the sign distribution of V_{1k} elements had no observable effect on the lineshapes of each probability.

This project gave me a sense of the assumptions underlying the theoretical model of quantum mechanical relaxation, and how quantum mechanical relaxation might behave in a real system wherein N_k and ΔE are severely limited. I can see that several deviations from the ideal behavior are possible, although the general exponential decay of the probability of the initial state being occupied, and logarithmic growth of the probability of the manifold being occupied, are accurate to a broad arrangement of situations.